CHANIRU WEERASINGHE

3Rd Year Undergraduate

INTERESTS

- **UI/UX design**
- **Web and Interesting tools making**
- **Software Development**
- **Exploring emerging technologies**

PROFILE SUMMARY

Motivated Software Engineering undergraduate with hands-on experience in full-stack web and mobile development. Skilled in React, Node.js, and Firebase, with a strong focus on creating user-friendly, scalable systems. Passionate about problem-solving, clean design, and continuous learning. Seeking a Software Engineering Internship to contribute to impactful projects and gain professional experience in modern development practices.

Contact

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Skills

- Programming Languages:

  Java, C, C#, JavaScript, HTML, CSS, SQL, PHP, PL/SQL, NoSQL query languages

- Frameworks / Libraries / Platforms:

  React, React Native, Node.js, Spring Boot, .NET, Tailwind CSS, Android SDK, jQuery

- Tools / Platforms / DevOps:

  Git, GitHub, VS Code, IntelliJ IDEA, Visual Studio, Android Studio, Firebase, AWS, MySQL, PostgreSQL, MongoDB, Docker, Postman, Linux / Windows, npm, Yarn, Maven, Gradle

- Soft Skills / Strengths:
  - Problem Solving / Analytical Thinking
  - Teamwork / Collaboration
  - Written Communication (**verbal limited due to stammering problem I've been having since my childhood**)
  - Critical Thinking / Decision Making
  - Creativity / Innovation / Design / UI/UX
  - Leadership / Focus & Persistence
  - Time Management / Prioritization
  - Detail-oriented, Sharpened Eye for Design & CSS

EDUCATION

BSc(Hons) Computing **Coventry University (UK)**	**2025-2027 (Present)**
Higher National Diploma in Software Engineering	**2024-2025**
Diploma in Software Engineering	**2023-2024**
Certificate in Computer Science	**2022-2023**

Projects / Side Projects

1. River Range Resort Pvt(LTD) Hotel Website Guest Portal (Side Project)

Full-stack guest portal built with React.js (v18) and Firebase, featuring an enterprise-grade booking engine with real-time inventory synchronization.

- Integrated Google Gemini Pro & OpenAI to power an AI Concierge that provides instant guest assistance and dynamic route summarization.
- Implemented a high-fidelity, luxury UI using GSAP and Framer Motion for premium cinematic transitions and micro-animations.
- Developed secure payment workflows integrated with the **DFCC Bank IPG** (Internet Payment Gateway) and support for LankaQR/Mobile Wallets.
- Utilized Google Maps API for custom location-based services and route visualization for arriving guests.

Designation: **Lead Developer & System Administrator (River Range Resort Pvt Ltd)**
https://riverrangehotel.com/

2. River Range Resort Pvt(LTD) Hotel Web Application Staff ERP System (Side Project)

Internal ERP & Management System built with React.js and Node.js/Express to centralize luxury resort operations, guest CRM, and financial settlements.

- Architected a "Zero Trust" Security Model featuring Static IP lockdown and Secure Mesh VPN access to protect sensitive financial data.
- Developed a Reservation Lifecycle Management module with 1:1 verified availability workflows to automate check-ins and prevent overbooking.
- Engineered a Financial Control Center for real-time tracking of DFCC Bank transactions, daily revenue analytics, and automated settlement reporting.
- Created a custom CMS Integration that allows staff to manage pricing, room availability, and media on the public website without developer intervention.

Designation: **Lead Developer & System Administrator (River Range Resort Pvt Ltd)**
https://#########.riverrangehotel.com/

Web Tools

3) V-Frame: High-Performance Scroll Video Engine (Side Project)

A web-based tool for extracting image sequences from videos, optimized for building high-quality scroll animations and scrollytelling websites.

- Built a Node.js & FFmpeg engine to extract high-quality WebP, PNG, and JPG frame sequences for web animations.
- Implemented multi-threaded processing to handle thousands of frames with hardware-accelerated conversion speeds.
- Developed real-time progress tracking via Server-Sent Events (SSE) for live extraction and ZIP generation updates.
- Integrated automated cleanup to purge temporary files post-download, ensuring server efficiency and security.
- Designed a premium glassmorphic UI with smooth scrolling (Lenis) and interactive canvas backgrounds.

4) Sonic Pulse – High-Performance Audio Extraction Engine (Side Project)

Full-stack media processing application built with React.js and a containerized Node.js backend, featuring a high-fidelity MP3 conversion engine with real-time audio visualization.

- Built a hybrid React/Node.js stack using Docker to offload heavy FFmpeg and high-bitrate extraction workloads.
- Integrated Web Audio API to develop a real-time frequency visualizer that syncs live audio data with the UI.
- Engineered a secure audio pipeline with server-side validation to ensure 320kbps studio-quality output.
- Designed a cinematic UI featuring glassmorphism and fluid micro-animations for a premium user experience.

5) TaskTrace Premium: Real-Time Project Management Platform (Side Project)

A premium Jira-style platform for real-time team collaboration, project tracking, and performance analytics with a focus on high-end aesthetics.

- Architected a multi-page glassmorphic UI using Vanilla JS and Tailwind CSS, ensuring full responsiveness and a state-of-the-art user experience.
- Integrated Firebase Realtime DB and Chart.js to deliver live data syncing and dynamic analytics for sprint velocity and task burndown tracking.
- Engineered a custom state management system with secure authorization layers and a dynamic theme engine that adapts to project metadata.

6) CutOut AI – Professional-Grade AI Background Removal & Object Detection (Side Project)

A real-time image processing tool using Vision Transformers (ViT) for automated background removal and object classification.

- Developed a Python/FastAPI backend supporting multi-engine AI segmentation (ISNet & BiRefNet) for professional-grade precision.
- Engineered a binary streaming pipeline to deliver high-transparency PNGs and object metadata within a single, high-performance request cycle.
- Designed a cinematic glassmorphic UI with Next.js 15 and Framer Motion, optimized for complex micro-interactions and real-time feedback.

7) DraftPoint: Next-Generation Portfolio & Project Receiveing Hub (Side Project)

A premium web platform designed for high-end developers to automate client onboarding through dynamic project estimation and an interactive requirement gathering workflow.

- Built a real-time project cost estimator using React state logic to dynamically calculate development budgets based on service tiers, feature selections, and recurring maintenance cycles.
- Developed a custom tag-based requirement collector using Framer Motion to instantly transform text inputs into interactive visual chips, streamlining the project scoping workflow.
- Integrated a serverless email pipeline via Vercel Functions and Nodemailer to securely automate project request submissions and client lead capture directly from the UI.
- Engineered a dynamic lead generation tool that programmatically builds personalized WhatsApp consultation drafts based on user-selected technology tokens.
- Designed a premium glassmorphic UI system leveraging Tailwind CSS 4 and physics-based motion transitions to ensure a high-performance, responsive experience across all devices.

Web Tools

8) OrderBy: High-Performance Client-Side Image Processing & Organization Engine (Side Project)

A premium web-based utility for bulk image renaming, format conversion, and structural organization, leveraging local browser resources for privacy-first, serverless processing.

- Built a high-performance image processing engine using the HTML5 Canvas API to perform bulk renaming and format conversions (WebP, PNG, JPG) entirely on the client side with zero server dependency.
- Implemented a "Multi-Lane" parallel processing architecture that utilizes browser hardware concurrency to handle multiple image workloads simultaneously, significantly reducing batch execution time.
- Developed a virtual folder organization system allowing users to categorize images into custom groups that are dynamically compiled into a structured, downloadable ZIP file using JSZip.
- Designed a premium, glassmorphic UI featuring smooth Framer Motion animations, real-time processing progress tracking, and a modern design system built with Tailwind CSS 4.

9) Studio AI: Browser-Based Batch Image Processor & On-Device AI Engine (Side Project)

A high-performance web tool designed for privacy-focused batch image editing, enabling automated background removal and intelligent resizing directly within the user's browser.

- Built a client-side execution engine using WebAssembly (WASM) to run complex AI models locally, ensuring 100% data privacy and eliminating server-side processing latency.
- Developed a batch processing pipeline using JSZip and FileSaver.js to handle concurrent image operations for hundreds of files and export results as optimized ZIP archives.
- Engineered an interactive canvas preview system with real-time zoom and positioning controls, supporting automated "Cover", "Contain", and "Stretch" scaling modes.
- Implemented a performance-tuned UI featuring CSS Custom Properties for dynamic theme switching and optimized asset loading for a minimal initial landing footprint.
- Integrated sophisticated format conversion logic to allow instant export across PNG, JPG, and WEBP formats with configurable compression quality settings.

10) NeuralScale – Browser-Native AI Image Upscaling Engine (Side Project) - Ongoing

A high-performance web tool designed to upscale and enhance images using local neural engines. It performs all computations on the user's hardware, ensuring 100% privacy and eliminating the need for server-side processing.

- Built a client-side execution engine using ONNX Runtime Web to run super-resolution models directly in the browser via WebGPU.
- Implemented a smart tiling algorithm to process large images by segmenting them into patches, preventing VRAM overflow and browser crashes.
- Developed a batch processing pipeline to handle multiple image workloads with automated queue management and real-time inference timing.
- Engineered a high-fidelity comparison preview system using an interactive slider to visualize AI-enhanced details against the original input.
- Designed a modern glassmorphic interface using Tailwind CSS 4 and Framer Motion, focusing on a minimal and intuitive user experience.

Hosted Websites

- https://riverrangehotel.com/
- https://river-range-resort.web.app/ (going to be hosted on the main domain)
- https://mingxing-10d5f.web.app/
- https://la-cherie-bakery.web.app/
- https://tasktrace-todo.web.app/
- https://draft-point.vercel.app/
- https://chanii2024.github.io/studio-ai/
- https://orderby-uploads.web.app/
- https://chanii2024.github.io/turbo-compress/

Past Projects / Experience

11. Auralia Streaming App

Full streaming app (similar to Spotify) using Java, IntelliJ, Tomcat, and Maven, integrating audio/video playback, user authentication, and personalized recommendations.

12. SCMS-Project-Frontend (Smart Campus Monitoring System)

Frontend for Smart Campus Monitoring System (second-year final project) using React and CSS with real-time data connection.

13. Tution Management Application

Java-based student management and educational streaming platform. Features a Spotify-style UI, Android integration, and Firebase-powered real-time updates, secure authentication, and personalized content recommendations.

14. Urbanfood Restaurant Website

Frontend for food delivery platform using Java with SQL and NoSQL, integrating menus, orders, and tracking.

15. LibriSense-IoT

Smart library IoT system using sensors and real-time dashboard to track book usage and seating occupancy, enabling efficient library management.

16. VisionGateOptometrist

Web application using C# in Visual Studio for optometry appointments and patient records, including all relevant components for scheduling and management.

17. Auralia User Service

Single microservice for Auralia streaming platform (EAD2 module) handling user authentication and profile management using Node.js.

18. Machine Learning - Fetal Health Classification

Machine learning project for coursework involving data preprocessing, model training, and evaluation using Python and scikit-learn.

19. Auralia Movie Recommendation

Movie recommendation system using data structures and algorithms (PDSA module) to provide personalized suggestions.

References

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